

Heat

Atoms and molecules are always jiggling around. The faster they move, the more energy an object has. We feel this energy as heat. When something is hot, its atoms are moving quickly. When something is cold, its atoms are moving slowly.

Feel the heat

Heat always tries to spread from hot things to cooler things. When you touch a hot object, heat energy flows into your skin, triggering sense cells that make your skin feel hot. When you touch a cold object, heat flows out of your skin, triggering a different feeling.



Temperature

The temperature of an object tells you how hot it is on a numbered scale. A device called a thermometer is used to measure temperature.



Sources of heat

Heat can be produced in several different ways.



Friction (rubbing) makes heat. If you pull on a rope, your hands will feel warm.



Combustion means burning. When something burns, it produces heat.



Electricity is used to create heat in electric ovens and heaters.

Warm glow

Heat escapes from warm objects as invisible rays that travel like light. We call this infrared radiation. Special cameras use infrared rays rather than light to take photos. Hot areas appear white or red and cold areas, such as this ice pop, appear black.

Keep your cool

Heat travels from the Sun as infrared rays. Just like light, infrared rays are reflected away by white objects but absorbed by black objects. In hot countries, people paint houses white to reflect the heat and keep the indoors cool.